

# Sam Weldon

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## Education

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### Columbia University

Dec 2026

*Bachelor of Arts in Information Science* GPA: 4.1/4.3

- **Coursework:** Advanced Programming, Advanced Systems Programming, Artificial Intelligence, C++ for C Programmers, Databases, Data Structures, Ethical & Responsible AI, Machine Learning & AI for Climate Science
- **Honors:** 4x Dean's List, Upsilon Pi Epsilon (UPE) Honor Society, GS Honor Society

### Housatonic Community College

May 2023

*Associate of Arts in History* GPA: 4.0/4.0

- **Honors:** Summa Cum Laude, Phi Theta Kappa Honor Society

## Experience

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### Bauplan Labs New York, NY

June 2026 – Present

*Software Engineer*

- Developed a git-for-data Apache Iceberg catalog in Rust and PostgreSQL, providing branching, merging, and time-travel over data lakehouse tables at a scale of millions of commits.
- Implemented REST APIs compatible with open-source industry standards, enabling third-party clients to work with the system unmodified.
- Designed relational database schemas for high-concurrency workloads, with careful attention to transaction isolation, locking strategies, and race-condition safety.
- Wrote design documents and led technical discussions on system architecture, including trade-off analyses of garbage collection strategies, conflict-detection rules, and schema evolution.

### DAPLab at Columbia University New York, NY

[github.com/ElaineAng/db-fork](https://github.com/ElaineAng/db-fork)

*Research Assistant*

Jan 2026 – Present

- Designed and implemented benchmarking infrastructure in Python and SQL to evaluate performance of branchable database systems, including PostgreSQL, Neon, Xata, and Doltgres.
- Built parameterized experiments to measure throughput, latency, and storage overhead under varying concurrency levels and branching topologies.
- Co-authored a research paper submitted to academic journals: <https://arxiv.org/abs/2604.17180>.

### Columbia University New York, NY

Sept 2025 – Present

*Teaching Assistant*

- Served as a TA for the following courses: Advanced Programming, From Algorithmic Thinking to Development, C++ Deep Dive for C Programmers
- Led recitations and office hours for 300+ students, focusing on debugging, memory management, and systems-level program behavior in C and C++.
- Taught software engineering practices including Git workflows, Makefiles, and Linux-based development environments.
- Guided implementation of data structures, graph search and dynamic programming algorithms, and functional programming in C, C++, Java, and Python.

### Sing Sing Prison Museum Ossining, NY

[github.com/samweldon9393/SingSingPrisonMuseum-maps](https://github.com/samweldon9393/SingSingPrisonMuseum-maps)

*Software Engineering Fellow*

June 2025 – Aug 2025

- Built and deployed a production React and TypeScript application for an interactive museum exhibit.
- Designed modular frontend components and implemented state management for scalability and maintainability.
- Developed data visualization workflows in Python to generate animations from real-world datasets.

## Skills

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**Languages:** Rust, C++, C, Python, SQL

**Software Engineering:** Data Structures & Algorithms, Object-Oriented Programming, Functional Programming, Backend Development, Databases, Systems Programming, Concurrency Control (multi-threading, IPC), Linux Containers

**Technologies:** Linux, Bash, Git & GitHub, PostgreSQL, Pandas, AWS, Google Cloud, HTML, CSS, REST APIs